AUXILIARY & SULPHURIC ACID PLANT

MSFE HEATER CLEANING PROCESS

DEV SATISH
2023B1A40833P



MSFE HEATER - STRUCTURE & PURPOSE

The heater unit used in a Multiple-Effect Evaporator (MSFE) system. It is essentially a shell-and-tube heat exchanger used to heat the feed liquid before it enters each evaporator effect.

Purpose of an MSFE Heater:

An MSFE system uses steam to evaporate water or solvent from a feed liquid in multiple stages ("effects") to increase energy efficiency. The MSFE Heater plays a crucial role in this process.

1. Heats Feed Liquid:
   - Steam is passed through the shell side of the heater.
   - The feed liquid passes through the tube side.
   - Heat is transferred from the steam to the liquid.
2. Prepares for Evaporation:
   - The heated liquid is then sent to the evaporator body, where it boils under vacuum, separating water as vapor.
   - Heating is done just below the boiling point at reduced pressure, minimizing energy use.
3. Repeats Across Effects:
   - In later effects, vapor from the previous stage is reused as the heating medium, and a new heater is used in each stage.

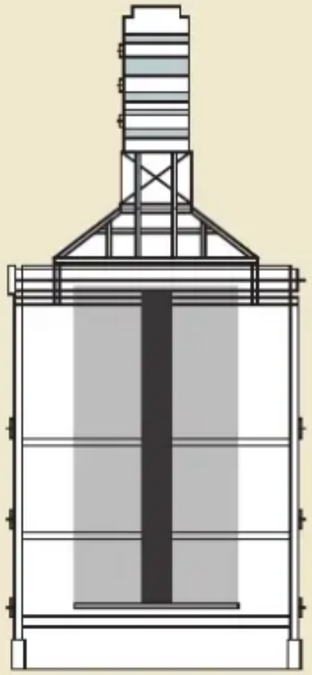
At Grasim Industries, especially in VSF and chemical recovery, MSFE heaters:

- Concentrate chemical effluents or weak acid streams
- Improve steam economy (reduce fresh steam use)
- Handle high-TDS, corrosive feed liquids, requiring durable materials (titanium)
- Are critical in minimizing wastewater volume and energy consumption

Control Measures adopted MSFE RO feed increased for more water recycling
> Water saving of 600 m3/day

Why is Cleaning Important?

Because the feed liquid can cause scaling and fouling inside the tubes, reducing heat transfer. That's why your project on Advancements in MSFE Heater Cleaning is important—it focuses on making cleaning faster, safer, and more efficient.







Ash and unburned hydrocarbons depostied in heaters convection tubes disturb effective heat transfer

MSFE PROJECT EQUIPMENT & PRODUCTS

Tube Holder Plates

- Function: These plates hold and align the tubes inside the heater shell.
- Use in MSFE: In shell-and-tube heat exchangers, they ensure that heat transfer tubes are securely positioned, allowing efficient steam heating of the feed solution.
- Material: Usually made of corrosion-resistant metals (e.g., SS 316) to withstand hot, corrosive liquids.

Heater Shells and Evaporators

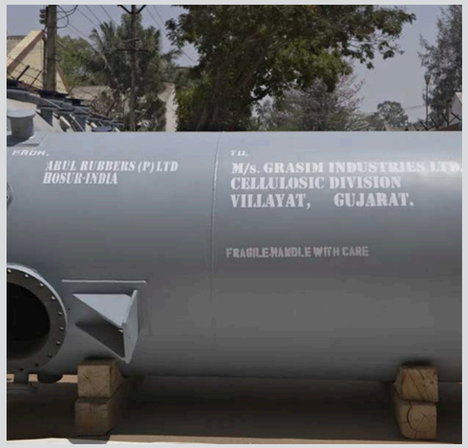
- Function: The heart of MSFE, where feed liquid is heated to cause evaporation.
- Heater Shells: Outer shell where steam is passed to heat the liquid inside tubes.
- Evaporators: Remove water from the feed by boiling it under vacuum, across multiple stages (effects), increasing steam efficiency.
- Material: Built to withstand thermal stress and corrosive environments.

Sand filter

- Function: Used for pre-treatment of feedwater or process liquid.
- Use in MSFE: Removes suspended solids that could cause scaling or fouling inside heater tubes and evaporators.
- Layers: Typically consist of sand, gravel, and pebbles to trap particles.



GRID PLATES



HEATER SHELL



SAND FILTERS



CYCLONE SEPERATOR

- Function: Use centrifugal force to separate vapors from entrained liquid droplets.
- Use in MSFE: After evaporation, these remove any liquid carryover from the vapor stream, ensuring dry steam moves to the next effect or for condensation.
- Efficiency: Prevents loss of product and scaling in downstream equipment.

MSFE PROJECT EQUIPMENT & PRODUCTS

Star Rollers

- Function: Typically part of a scraper or rotating drum mechanism.
- Use in MSFE: Helps in concentrated slurry handling or crystal discharge from final effects.
- Design: Star-like rotating components to push solids/slurry and prevent blockages.

Intel & Flash Nozzles

- Inlet (Intel) Nozzles: Introduce feed liquid into the evaporator at specific locations to maximize surface contact.
- Flash Nozzles: Inject partially heated feed into a lower pressure effect, causing flash evaporation.
- Goal: Improve phase change efficiency and reduce scaling by directing flow smartly.

Pipes & Fittings

- Function: Carry steam, condensate, feed liquid, and vapor between various components.
- Types: Includes elbows, tees, reducers, valves, etc.
- Material: Must be resistant to corrosion, pressure, and temperature changes (e.g., SS 304, SS 316, FRP-lined pipes).



STAR ROLLERS



FLASH NOZZLE



PIPES AND FITTINGS



GASKETS

- Function: Ensure leak-proof sealing between flanged connections, covers, and joints.
- Use in MSFE: Maintain vacuum integrity and steam-tight seals in heaters and separators.
- Materials: Often made from EPDM, Viton, or PTFE, selected based on chemical compatibility

RESEARCH PAPER 1

Aim :

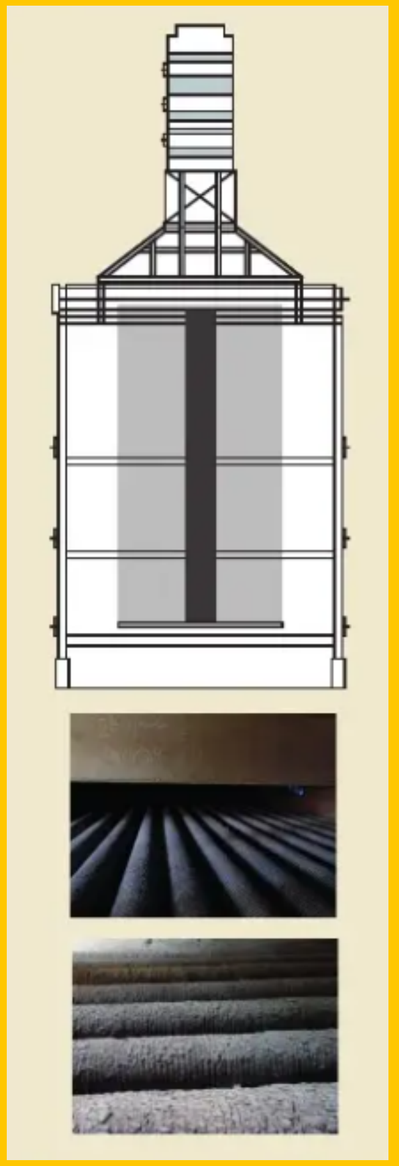
Cleaning Heater Convection Tubes

Background :

Fired heaters are critical in refineries, but over time, ash, unburned hydrocarbons, and foreign matter build up on the convection tubes, reducing heat transfer. The convection section is difficult to clean due to tight spacing, tube arrangement, and access issues. Liquid fuels cause more deposits than gas fuels, increasing the fouling risk.

Problems :

Narrow gaps between tubes (as little as 30 mm) make mechanical cleaning very hard. Tube bundles are long (>2 m), and some areas are inaccessible by hand. Earlier methods involved spraying water and chemicals, which were ineffective for hard-to-reach areas or stubborn deposits.



Ash and unburned hydrocarbons depostied in heaters convection tubes disturb effective heat transfer

RESEARCH PAPER 1 - ADVANCEMENTS

Advancements in Fired Heater Cleaning Techniques

1. Portable Cleaning Machines : Automated machines with multi-directional rails and high-pressure jets (100 bar) clean large tube surfaces efficiently without scaffolding, reducing manual effort and hazards.

2. Hand Jet Cleaning : Ultra-high-pressure (2700 bar) water jets with flexible nozzles are manually inserted to reach tightly packed or deep tube areas, ensuring thorough cleaning where machines can't reach.

3. Chemical Cleaning Integration : A combination of chemicals is used to dissolve hard deposits and enhance cleaning efficiency :

Hydrochloric Acid (1–2%) – removes scale and oxides.

Ammonium Bifluoride – for silica-based or stubborn deposits.

Surfactants/Detergents – emulsify hydrocarbons and improve cleaning of oily surfaces.

Corrosion Inhibitors (e.g., PDTDC) – protect metal surfaces during acid exposure.

Neutralizers (e.g., Sodium Carbonate) – stop acid reaction after cleaning.

Passivators (e.g., Citric or Nitric Acid) – restore protective oxide layer on metal.

4. Endoscopic Inspection : Industrial cameras are used before and after cleaning to verify effectiveness and guide re-cleaning if needed..

5. Water Quality Monitoring : Cleaning progress is tracked by observing water clarity (dirty to clear), helping avoid under- or over-cleaning.

6. Optimized Cleaning Sequence : A defined sequence (debris removal => chemicals => jets => rinse => inspection) improves safety, speed, and effectiveness.

7. Performance Monitoring : Post-cleaning analysis



Figure 3 Portable tube cleaning machine

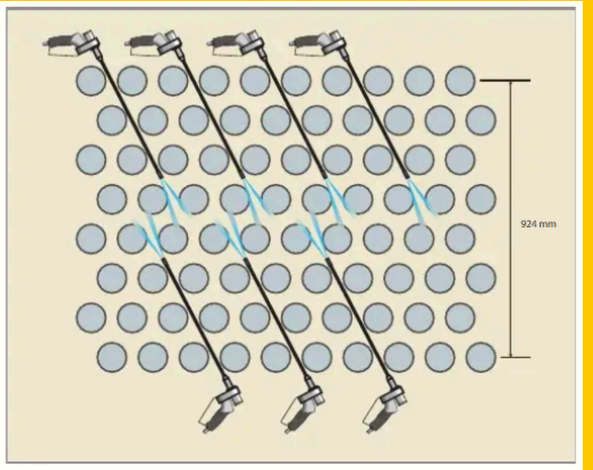


Figure 6 Hand jet cleaner

RESEARCH PAPER 2

Aim :

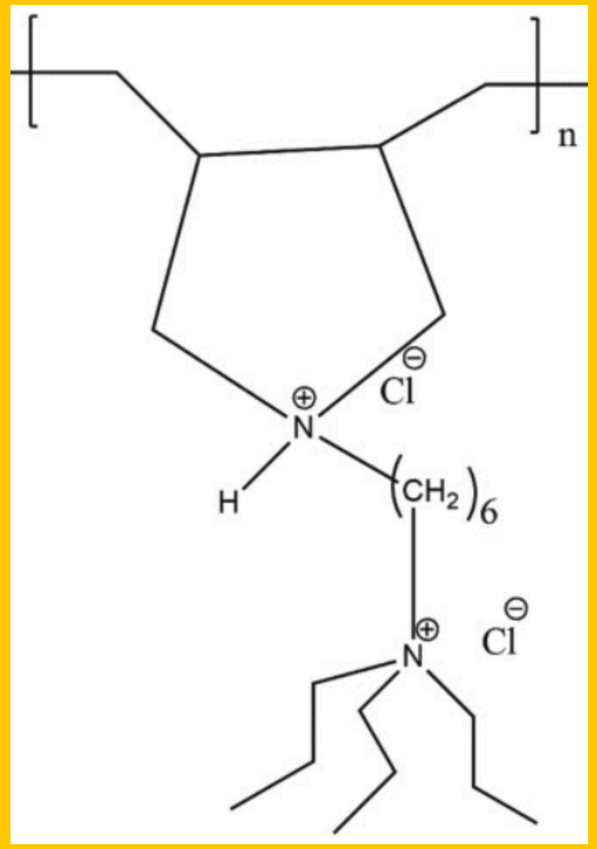
Desalination Plant Acid Cleaning: Corrosion & Inhibition Study

Background :

Elucidating the corrosion characteristics of brine heater and evaporator condenser alloys during acid cleaning of MSF plants and its mitigation. MSF plants suffer from scaling (mineral deposits) on brine heater and evaporator tubes. These scales are removed using acid cleaning, often with 3% HCl at 40 °C for 72 hours. But acid cleaning damages metal components, especially Cu–Ni alloys, unless corrosion inhibitors are used.

Experimental Setup :

Materials studied : CuNi10 (10% nickel), CuNi30 (30% nickel)
New corrosion inhibitor : PDTDC (a nitrogen-containing ionic polymer)
Goal : See if PDTDC can protect these metals under harsh acid cleaning conditions.



Chemical structure of poly (N1, N1-diallyl-N6, N6, N6-tripropylhexane-1,6-diaminium chloride) (PDTDC).

RESEARCH PAPER 2

Results :

1. Corrosion Behavior

Both alloys corrode significantly during acid cleaning. CuNi30 showed better resistance than CuNi10 due to higher Ni content. Corrosion led to pitting and surface roughness.

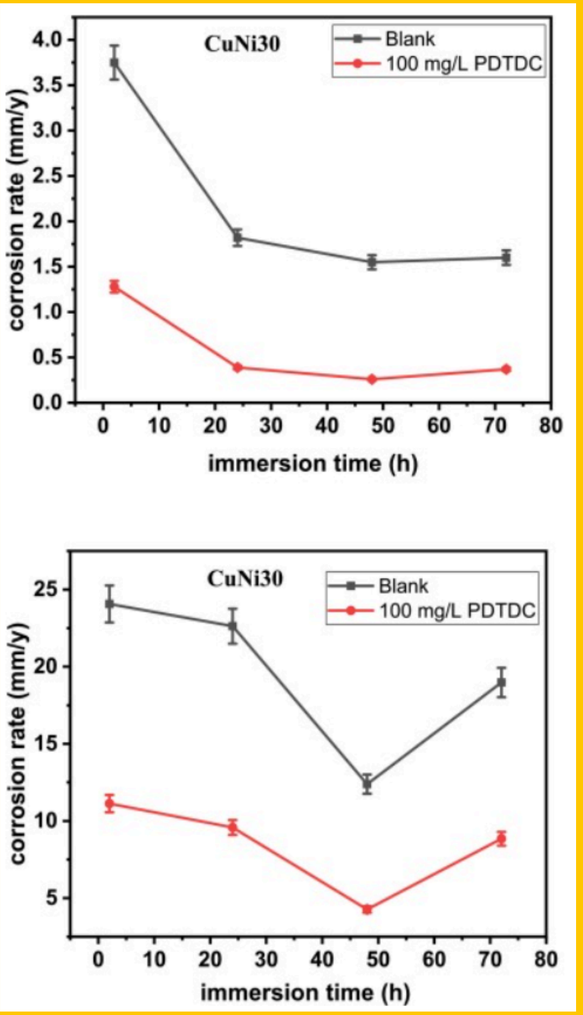
2. Effectiveness of PDTDC

PDTDC adsorbs onto the metal surface, mainly at cathodic sites. It blocks hydrogen ion (H^+) reduction, slowing down corrosion. It's a mixed-type inhibitor, but cathodic dominant.

3. PDTDC remains effective up to 72 hours and 50 °C, with best results at 48 hours.

4. PDTDC performs well under flow conditions due to improved contact with metal surfaces.

Alloys	Without PDTDC	With 100 mg/L PDTDC	Inhibition Efficiency
CuNi30	1.60 mm/year	0.37 mm/year	77%
CuNi10	18.96 mm/year	8.85 mm/year	53%



The variation of corrosion rate (mm/y) with immersion time (h) for the corrosion of CuNi30 and CuNi10 alloys in 3 wt% HCl solution at 30 °C.

RESEARCH PAPER 2

Conclusion :

PDTDC is a highly effective and stable corrosion inhibitor for acid cleaning in MSF desalination plants, showing superior performance with CuNi30 and reasonable effectiveness with CuNi10. It serves as a cost-efficient and promising alternative to conventional inhibitors and has potential for use in multi-metal corrosion protection formulations. The study holds strong industrial relevance, especially for desalination units, brine heaters, evaporator cleaning, and systems like MSFE that use Cu–Ni alloys in auxiliary or chemical plants.

$$\mathrm{Cu} \rightarrow \mathrm{Cu}^{+} + e^{-} \quad (oxidation\ at\ the\ metal/oxide\ interface)$$

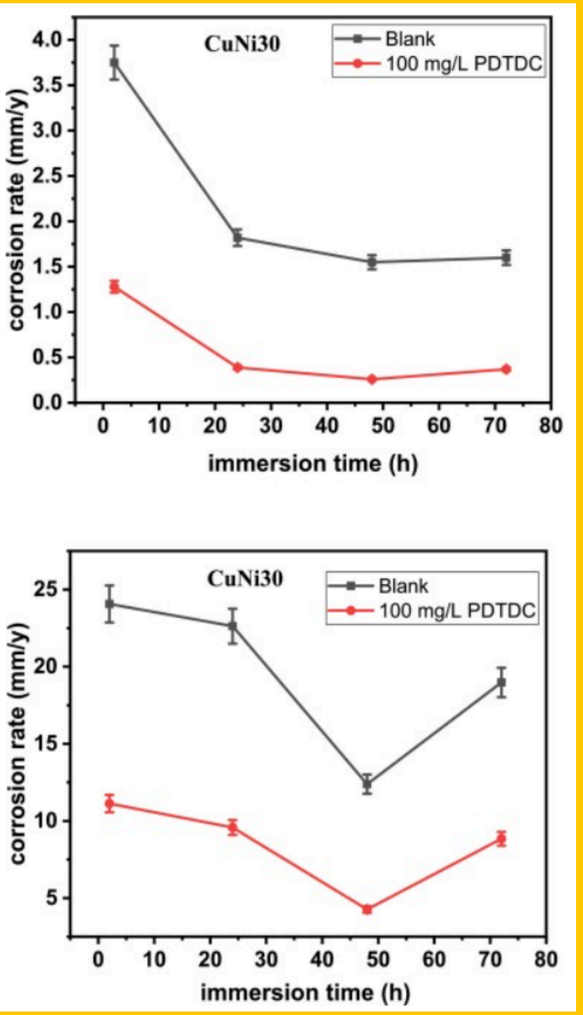
$$\mathrm{Cu}^{+}_{(metal\ /\ oxide)} \rightarrow \mathrm{Cu}^{+}_{(oxide\ /\ solution)} \quad (solid-state\ mass\ transport)$$

$$\mathrm{Cu}^{+} + \mathrm{Cl}^{-} \leftrightarrow \mathrm{CuCl}^{-}_{\mathrm{ads}} \quad (adsorption\ at\ the\ oxide/solution\ interface)$$

$$\mathrm{CuCl}^{-}_{\mathrm{ads}} + \mathrm{Cl}^{-} \leftrightarrow \mathrm{CuCl}^{-}_{2} \quad (desorption\ at\ the\ oxide/solution\ interface)$$

$$\mathrm{Cu} + 2\,\mathrm{Cl}^{-} \rightarrow \mathrm{CuCl}^{-}_{2} + e^{-}$$

$$2\mathrm{CuCl}^{-}_{2} + \mathrm{H_2O} \rightarrow \mathrm{Cu_2\,O} + 2\mathrm{H}^{+} + 4\,\mathrm{Cl}^{-}$$



The variation of corrosion rate (mm/y) with immersion time (h) for the corrosion of CuNi30 and CuNi10 alloys in 3 wt% HCl solution at 30 °C.

RESEARCH PAPER 3

Novel and robust machine learning approach for estimating the fouling factor in heat exchangers.

Aim :

Develop an accurate, reliable, and generalizable machine learning model to estimate the fouling factor (Rf) in heat exchangers based on operational parameters, in order to improve energy efficiency, reduce maintenance costs, and enable predictive cleaning.

Background :

Fouling in heat exchangers, including MSFE heaters, is the buildup of unwanted materials like scale or corrosion on surfaces, reducing heat transfer efficiency. The fouling factor (Rf) is a key performance metric representing thermal resistance caused by fouling.

Traditionally, Rf is estimated using simplified empirical formulas, but these often fail under dynamic and non-linear industrial conditions. The paper proposes using machine learning for accurate and robust Rf prediction to help industries optimize cleaning schedules and reduce energy loss.

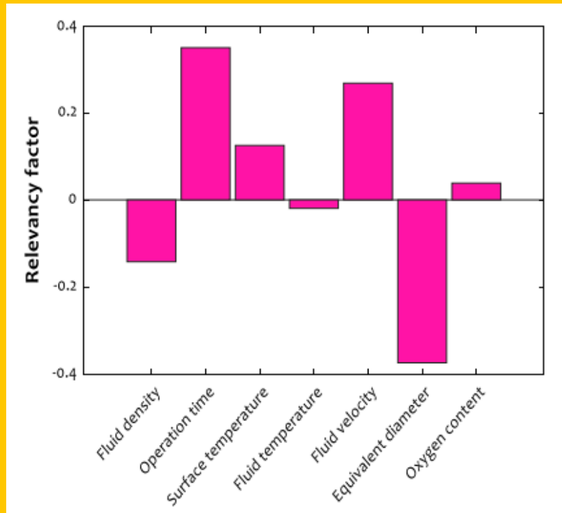


Fig. 2a. Relevancy factors between fouling factor and the corresponding features determined by the Pearson method.

- Positive value: as the feature increases, the fouling factor increases.
- Negative value: as the feature increases, the fouling factor decreases.

RESEARCH PAPER 3

Training Dataset

- Total samples: 11,626 fouling observations from prior lab work by Davoudi and Vaferi (2018)
- Features used:
  - Fluid velocity and density
  - Oxygen content
  - Surface and fluid temperature
  - Operation time
  - Pipe/equivalent diameter
  - Fluid composition

Methodology Overview

1. Preprocessing
   a. Z-score normalization applied to scale input features
   b. Non-linear transformation of the fouling factor — √Rf (square root) worked best for prediction

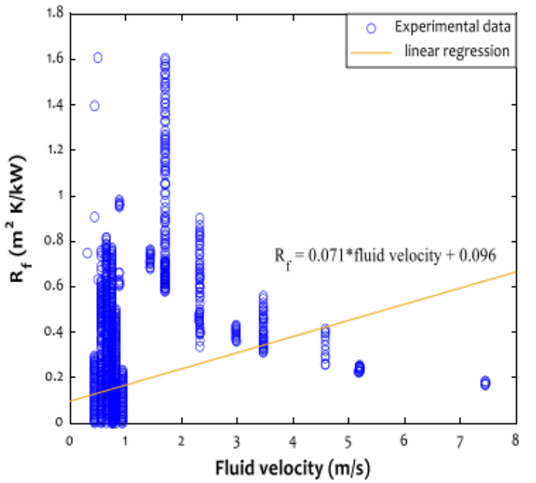


Fig. 2b. Variation of the fouling factor by the fluid velocity (without considering the effect of other parameters) and its associated linear regression.

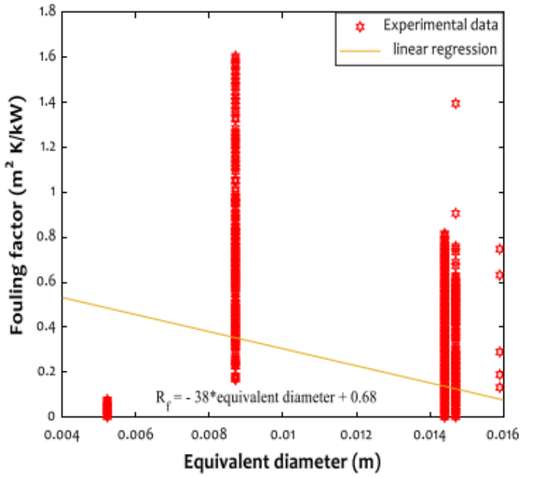


Fig. 2c. Variation of the fouling factor by the equivalent diameter (without considering the effect of other parameters) and its associated linear regression.

RESEARCH PAPER 3

Methodology Overview

1. Preprocessing
    a. Z-score normalization applied to scale input features
    b. Non-linear transformation of the fouling factor — √Rf (square root) worked best for prediction
2. Model Training
    a. Used 5-fold cross-validation (80% of data) to tune and validate models. Tested on a completely unseen test set (20% of data)
3. Models Evaluated:
    a. Gaussian Process Regression (GPR) (best performer)
    b. Decision Trees
    c. Bagged Trees
    d. Support Vector Regression (SVR)
    e. Linear Regression (baseline)
4. Evaluation Metrics:
    a. R^2 (coefficient of determination)
    b. MAPE (mean absolute percentage error)
    c. MAE, MSE, RAE (other error metrics)

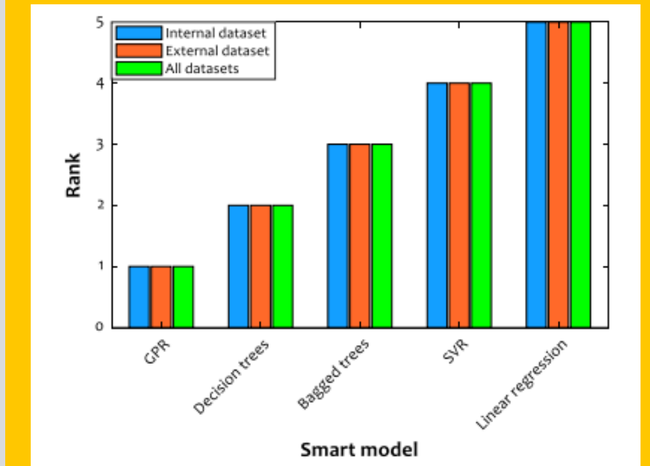


Fig. 3. The outcome of applying ranking analysis to the reported results in Table 4.

RESEARCH PAPER 3

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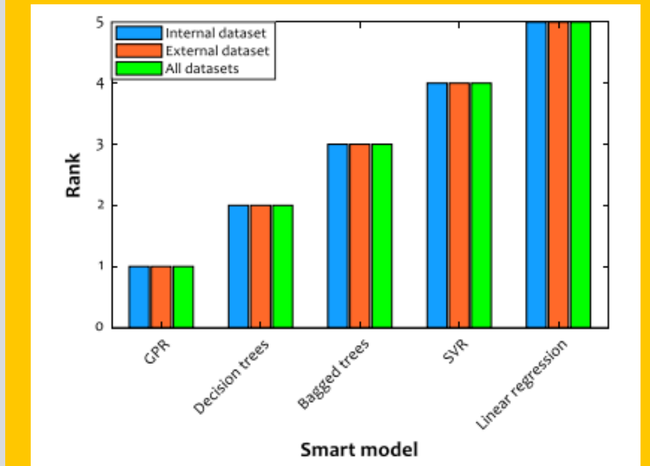


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RESEARCH PAPER 3

Real World Applications

- Can be used in:
  - Desalination (e.g., MSFE brine heaters)
  - Petrochemical and food processing plants
  - Power plant heat exchangers
- Helps predict when fouling builds up so cleaning can be scheduled proactively
  a. Reduces downtime, chemical use, and maintenance costs

Key Innovation

- The paper found that predicting √Rf instead of Rf significantly improved all ML models' generalization and accuracy.

Conclusion

This paper provides a validated, accurate, and deployable ML model (GPR) for predicting fouling in heat exchangers. It's a strong foundation for the MSFE heater cleaning process, especially proposing predictive maintenance using machine learning.

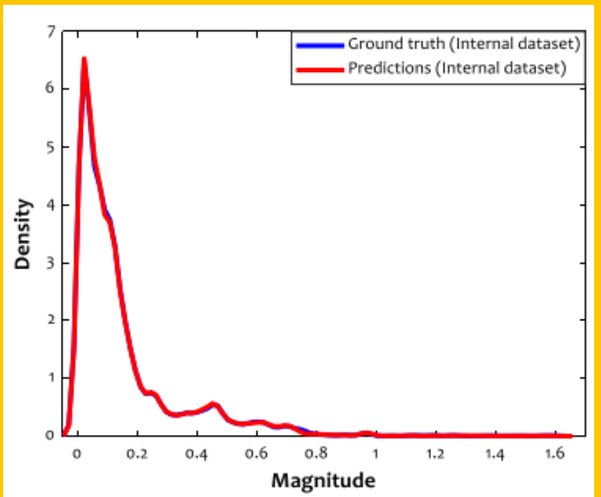


Fig. 7a. Kernel density estimation (KDE) plot for the internal datasets.

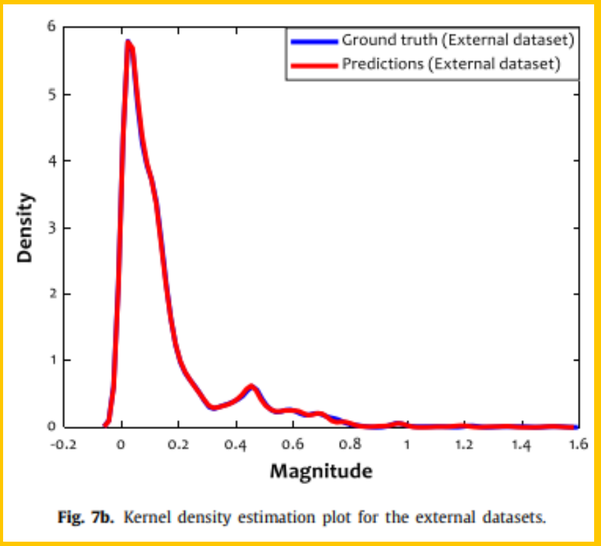


Fig. 7b. Kernel density estimation plot for the external datasets.

PROTOTYPE

Developed a real-time corrosion detection web application using ReactJS and TensorFlow.js, which leverages a deep learning model trained on image data to detect rust (corrosion) using a mobile or webcam feed. The app accesses the user's camera, classifies live video frames, and displays corrosion status with visual feedback. The prototype is lightweight, browser-based, and mobile-optimized for field-level inspection use.

Features Implemented

- Real-time prediction display (Rust Detected / No Rust)
- Corrosion detection using a pre-trained CNN model

Planned Features

- Pipe detection and classification using object detection models (e.g., SSD or YOLO)
- Diameter estimation using bounding box + pixel-to-mm conversion
- Fouling level prediction using ML regression models (e.g., GPR or LSTM) trained on MSFE operating data
- Image capture and export for maintenance logging or AI model re-training



Protype Preview

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THANK YOU